GATE-2007

BT: BIOTECHNOLOGY

Q.1 – 6 carry one mark each

1. The specific growth rate (μ) of a microorganism in death phase is
   (a) 0 (zero) (b) μ_{max}
   (c) less than zero (d) greater than zero
2. Which of the following reagents is used for harvesting anchorage-dependent animal cells from culture vessels ?
   (a) Trypsin/Collagensase
   (b) Trypsin/Collagen
   (c) Collagen/Fibronectin
   (d) DMSO
3. Protein binding regions of DNA are identified by one of the following techniques
   (a) finger printing
   (b) foot printing
   (c) southern blotting
   (d) western blotting
4. Plant secondary metabolites
   (a) help to increase the growth rate of plant
   (b) help in plant reproduction processes
   (c) provide defense mechanisms against microbial attack
   (d) make the plant susceptible to unfavorable conditions
5. Si RNA(s) interfere at
   (a) transcriptional level
   (b) post-transcriptional level
   (c) DNA replication level
   (d) translational level
6. Presence of $CX_{2-4}CX\phi X_8HX_3H$ sequence in a protein suggest that it is
   (a) a protein kinase
   (b) GTP binding protein
   (c) zinc finger protein
   (d) lipase

Q. 7 – 24 carry two marks each

7. A protein binds to phosphocellulose column at pH 7.0 and elutes at pH 8.0. If the protein has to be further purified on a DEAE Sephacel column, the binding buffer should have a pH of
   (a) 5 (b) 6
   (c) 7 (d) 8
8. Oils rich in PUFA are NOT desirable for bio-diesel production because
   (a) they form epoxides in presence of oxygen
   (b) they do not form epoxides in presence of oxygen
   (c) they have high ignition temperature
   (d) they solidify at low temperature
9. Gynogenesis is a process of development of haploid plants
   (a) from a fertilized cell of female gametophyte
   (b) from an unfertilized cell of female gametophyte
   (c) from isolated pollen grains
   (d) by selective elimination of chromosomes following distant hybridization
10. Match items in group 1 with correct examples from those in group 2

Group 1	Group 2
P. Catabolic product	1. Griseofulvin
Q. Bioconversion	2. Bakers yeast
R. Biosynthetic product	3. 6-Aminopenicillanic acid
S. Cell mass	4. Ethanol

   (a) P-4, Q-3, R-2, S-1
   (b) P-3, Q-4, R-1, S-2
   (c) P-4, Q-3, R-1, S-2
   (d) P-1, Q-4, R-3, S-2
11. A bioremedial solution to reduce oxides of nitrogen and carbon in flue gases is to integrate flue gas emission to
   (a) micro-algal culture
   (b) fish culture
   (c) mushroom culture
   (d) sericulture
12. The respiratory coefficient for the reaction
   $aCH_mO_n + bO_2 + cNH_3 \rightarrow dCH_\alpha O_\beta N_\gamma + eH_2O + fCO_2$
   is defined as
   (a) f/a
   (b) e/b
   (c) b/f
   (d) f/b

13. Match the methods available on world wide web in group 1 for performing the jobs listed in group 2

Group 1	Group 2
P. Boxshade	1. Searching family data base
Q. BCM launcher	2. Finding alignments
R. Prosite	3. Displaying alignments
S. PSI-BLAST	4. Searching for multiple alignments

(*a*) P-1, Q-3, R-2, S-4
(*b*) P-2, Q-3, R-2, S-4
(*c*) P-3, Q-4, R-1, S-4
(*d*) P-3, Q-2, R-1, S-4

14. Match the recombinant products in group 1 with their therapeutic applications in group 2

Group 1

P. Human growth hormone
Q. Platelet growth factor
R. Factor VIII
S. Erythropoietin

Group 2

1. Pituitary dwarfism
2. Chemotherapy inducred thrombocy-topenia
3. Haemophilia
4. Anaemia associated with chronic renal failure

(*a*) P-1, Q-2, R-3, S-4
(*b*) P-2, Q-1, R-3, S-4
(*c*) P-1, Q-4, R-3, S-2
(*d*) P-2, Q-4, R-3, S-1

15. Mobile genetic elements present in human genome are

P. long interspersed elements (LINEs)
Q. short interspersed elements (SINEs)
R. P elements
S. IS elements

(*a*) Q, R (*b*) P, Q
(*c*) P, R (*d*) Q, S

16. Match the following marker genes in group 1 with suitable selecting agent in group 2

Group 1	Group 2
P. npt II	1. Glyphosate
Q. aroA	2. Phosphinothricin
R. hpt	3. Kanamycin
S. bar	4. Hygromycin B

(*a*) P-1, Q-2, R-4, S-3
(*b*) P-3, Q-2, R-4, S-1
(*c*) P-2, Q-3, R-4, S-1
(*d*) P-3, Q-1, R-4, S-2

17. Determine the correctness or otherwise of the following Assertion [a] and Reason [r]

Assertion : Enzymatic method of tissue dispersion is milder than chemical and mechanical methods.

Reason : Enzymes work at optimal temperature and pH

(*a*) Both [a] and [r] are true and [r] is the correct reason for [a]
(*b*) Both [a] and [r] are true but [r] is not the correct reason for [a]
(*c*) [a] is true but [r] is false
(*d*) [a] is false but [r] is true

18. Match each parameter in group 1 with the appropriate measuring device in group 2

Group 1	Group 2
P. Pressure	1. Photometer
Q. Foam	2. Rotameter
R. Turbidity	3. Diaphragm gauge
S. Flow rate	4. Rubber sheathed electrode

(*a*) P-3, Q-4, R-1, S-2
(*b*) P-1, Q-3, R-2, S-4
(*c*) P-4, Q-1, R-2, S-3
(*d*) P-1, Q-2, R-3, S-4

19. Main functions of baffles in a bioreactor are

P. to prevent a vortex
Q. to increase aeration
R. to reduce interfacial area of oxygen transfer
S. to reduce aeration rate

(*a*) P, Q (*b*) Q, R
(*c*) R, S (*d*) P, S

20. How many kilograms of ethanol is produced from 1 kilogram of glucose in ethanol fermentation?

(*a*) 2.00 (*b*) 0.20
(*c*) 0.51 (*d*) 0.05

21. Meristems escape virus invasion because

(*a*) vascular system is absent in the meristem
(*b*) of low metabolic activity in the meristem
(*c*) the 'virus inactivating system' has low activity in the meristem
(*d*) of low endogenous auxin level

22. Downstream processing of an industrial process yielded a highly purified bioactive protein. This protein was subjected to cleavage by trypsin. Chromatographic separation of products resulted in 4 peptides (P, Q, R, S) with the following amino acid sequences

P. phe-val-met-val-arg
Q. ala-ala-try-gly-lys

R. val-phe-met-ala-gly-lys

S. phe-gly-try-ser-thr

Chemical cleavage of the same protein with cyanogenbromide and chromatographic separation resulted in three peptides (*i*, *ii*, *iii*) with the following sequences

(*i*) ala-gly-lys-phe-gly-try-ser-thr

(*ii*) ala-ala-pry-gly-lys-phe-val-met

(*iii*)val-arg-val-phe-met

The order of the peptides that gives the primary structure of the original protein is

(*a*) P, Q, R, S (*b*) Q, P, R, S

(*c*) Q, R, P, S (*d*) R, Q, P, S

Common Data for Questions 23. 24 :

Enzyme X converts substrates S_1 and S_2 (which are similar but not identical) to products P_1 and P_2 respectively

23. K_m values of enzyme X for substrate S_1 and S_2 are 0.1 mM and 0.01 mM, respectively This suggest that

P. enzyme X has more affinity towards S_1

Q. enzyme X has low affinity towards S_1

R. enzyme X has more affinity towards S_2

S. enzyme X has low affinity towards S_2

(*a*) P, Q (*b*) R, S

(*c*) Q, S (*d*) Q, R

24. What would happen if enzyme X is incubated with a mixture of 0.1 mM of S_1 and S_2?

(*a*) Products P_1 and P_2 are produced at equal concentrations

(*b*) Only product P_2 is produced

(*c*) More P_2 and less P_1 are produced

(*d*) More P_1 and less P_2 are produced

Linked Answer Questions: Q. 25 to Q. 28 carry two marks each

Statement for Linked Answer Questions 25 and 26 :

In a Fed-batch culture glucose solution is added with a flow rate of 2 m^3/day. The initial volume of the culture is 6m^3.

25. The volume of culture at the end of second day (neglect loss due to vaporization) is

(*a*) 6 m^3 (*b*) 8 m^3

(*c*) 10 m^3 (*d*) 12 m^3

26. What would be the dilution rate of the system at the end of second day?

(*a*) 2.00 (*b*) 0.20

(c) 0.02 (*d*) 0.01

Statement for Linked Answer Questions 27 & 28 :

Absence of cellulosic cell wall, high β-carotene content and GRAS status make *Dunaliella salina* a good model system for producing edible vaccines. 10^9 Cells of D. *salina* were electroporated with a high expression DNA vector containing an antigenic gene.

27. If 10^3 cells survived after electroporation, how many cells were killed during this process (round of to the nearest number) ?

(*a*) 10^9 (*b*) 10^8

(*c*) 10^6 (*d*) 10^5

28. The antigen is expressed as transmemb-rane protein with a single epitope on its extracellular domain. The cells that survived (assume 100% transfection and expression of protein) were incubated with a radio labeled Fab fragment (specific activity: 100 cpm/picomole) against this epitope. After washing, the cell pellet has 1000 cpm. The average number of epitopes present on a single recombinant alga are

(*a*) 6×10^9 (*b*) 1×10^9

(*c*) 6×10^3 (*d*) 1×10^6

ANSWERS

1. (*a*)	**2.** (*a*)	**3.** (*b*)	**4.** (*c*)	**5.** (*b*)	**6.** (*c*)	**7.** (*c*)	**8.** (*a*)	**9.** (*b*)	**10.** (*c*)
11. (*a*)	**12.** (*d*)	**13.** (*d*)	**14.** (*a*)	**15.** (*b*)	**16.** (*d*)	**17.** (*b*)	**18.** (*a*)	**19.** (*a*)	**20.** (*c*)
21. (*a*)	**22.** (*c*)	**23.** (*d*)	**24.** (*c*)	**25.** (*c*)	**26.** (*b*)	**27.** (*a*)	**28.** (*a*)		

EXPLANATIONS

1. Specific growth rate is defined as the increase in cell mass per unit time. In death phase, no cellular growth is observed. During this, the cell population continuously decreases with time and the population is said to have entered into death phase of the growth cycle. Therefore the value of specific growth constant for death phase is zero.

2. Cells are harvested when the cells have reached a population density which suppresses growth. For adherent cultures, we use trypsin or collagenase. Trypsin, collagenase, or pronase, usually in combination with EDTA, causes cells to detach from the growth surface, thereby aiding in harvesting of anchorage dependent animal cells from culture vessels.

3. DNA footprinting is a method of investigating the sequence specificity of DNA-binding proteins in vitro. This technique can be used to study protein-DNA interactions both outside and within cells. The regulation of transcription has been studied extensively, and yet there is still much that is not known. Transcription factors and associated proteins that bind promoters, enhancers, or silencers to drive or repress transcription are fundamental to understanding the unique regulation of individual genes within the genome. Techniques like DNA footprinting will help elucidate which proteins bind to these regions of DNA and unravel the complexities of transcriptional control.

4. Secondary metabolites often play an important role in plant defense against herbivory and other interspecies defenses. Humans use secondary metabolites as medicines, flavorings, and recreational drugs.

5. RNA interference (RNAi) also called post transcriptional gene silencing (PTGS), is a biological process in which RNA molecules inhibit gene expression, typically by causing the destruction of specific mRNA molecules. Two types of small ribonucleic acid (RNA) molecules: microRNA (miRNA) and small interfering RNA (siRNA) - are central to RNA interference.

6. A zinc finger protein is a DNA-binding protein domain consisting of zinc fingers ranging from two in the *Drosophila* regulator ADR1, the more common three in mammalian Sp1 up to nine in $TF_{111}A$. They occur in nature as the part of transcription factors conferring DNA sequence specificity as the DNA-binding domain.

7. If the protein is most stable at pH's above its pI, then an anion-exchange gel matrix should be used. If it is most stable below its pI, then a cation-exchange gel matrix should be used. Here the protein eluted at pH 8 so it would take pH7 to bind in DEAE column.

8. PUFAs are long-chain fatty acids containing two or more double bonds.

   They provide structural & functional characteristics, and are involved in a wide range of biological components including membranes (in phospholipids). Equally important PUFAs serve as precursors for conversion into metabolites that regulate critical biological functions.Biodiesel is a non-petroleum and environmentally-friendly alternative to petroleum-based diesel fuel and not fatty acid.PUFA is fatty acid it is not used as it forms epoxide in presence of oxygen.

9. A form of asexual reproduction related to parthenogenesis is gynogenesis. Here, offspring are produced by the same mechanism as in parthenogenesis, but with the requirement that the egg merely be stimulated by the presence of sperm in order to develop. However, the sperm cell does not contribute any genetic material to the offspring. Since gynogenetic species are all female, activation of their eggs requires mating with males of a closely related species for the needed stimulus.

10. Bioconversed product is 6 Amiono penicillinic Acid , Grisseofuvin is biosynthetic product, Bakers yeast have definite cell mass ie Biomass and Ethanol is a catabolic product.

11. The advantages of using a microalgal-based system in integration with flue gas emission are that:

    - High purity CO_2 gas is not required for algal culture. Flue gas containing varying amounts of CO_2 can be fed directly to the microalgal culture. This will simplify CO_2 separation from flue gas significantly.
    - Some combustion products such as NOx or SOx can be effectively used as nutrients for microalgae. This could simplify flue gas scrubbing for the combustion system.
    - Microalgae culturing may yield high value commercial products. Sale of these high value products can offset the capital and the operation costs of the process.
    - The envisioned process is a renewable cycle with minimal negative impacts on environment.

12. The respiratory quotient (RQ) is calculated from the ratio RQ = CO_2 eliminated / O_2 consumed where the term "eliminated" refers to carbon dioxide (CO_2) removed ("eliminated") from the body. So, RQ for the equation is f/b.

13. • **BOXSHADE** is a program for pretty-printing multiple alignment output. The program itself doesn't do any alignment, you have to use a multiple alignment program like ClustalW or Pileup and use the output of these programs as input for BOXSHADE. Of course, you can also use manually aligned sequences (in a proper format).

- **BCM Launcher**, i.e. Baylor College of Medicine Search Launcher is an on-going project to organize molecular biology-related search and analysis services available on the WWW by function by providing a single point-of-entry for related searches (e.g., a single page for launching protein sequence searches using standard parameters). It helps in finding alignments.
- **PROSITE** is a database of protein families and domains. It is based on the observation that, while there is a huge number of different proteins, most of them can be grouped, on the basis of similarities in their sequences, into a limited number of families. Proteins or protein domains belonging to a particular family generally share functional attributes and are derived from a common ancestor. PROSITE currently contains patterns and profiles specific for more than a thousand protein families or domains. Each of these signatures comes with documentation providing background information on the structure and function of these proteins.

14. HGH is used in dwarfism, Erythropoetin used in Annemia treatment, Factor VIII is been used in Haemophilia treatment and Plalet derived growth factor used in chemotherapeutic drug induced thrombocytopenia treatment.

15. Mobile genetic elements are the nucleotide sequences that can change their position in the genome. There are two categories of mobile genetic elements: DNA mobile genetic elements or DNA transposons and retroelements.

There are 3 kinds of retroelements:

- LINE(Long Interspersed Nuclear Elements)
- SINE(Short Interspersed Nuclear Elements)
- LTR(retrovirus like elements with Long Terminal Repeats)

16.

Marker Gene	*Enzyme encoded*	*Selective agent(s)*
hpt	*Hygromycin phosphotransferase*	*Hygromycin B*
npt II	*Neomycin phosphotransferase*	*Kanamycin*
aro A	*EPSP synthase*	*Glyphosate (Roundup)*
bar	*Phosphinothricin acetyltransferase*	*Phosphinothricin (bialaphos)*

17. A : Enzymatic method is mild. Chemical and mechanical methods may tend to destroy the tissue.

R : Enzymes show their catalytic activity at a particular optimum p H and temperature.

18. • The **Diaphragm Pressure Gage** uses the elastic deformation of a diaphragm (i.e. membrane) instead of a liquid level to measure the difference between an unknown pressure and a reference pressure.

- In an antifoam system; if the foam contacts a **rubber sheathed electrode**, an electric unit actuates a solenoid valve to allow the passage of a sterile antifoam agent into the fermentor.
- A final widely used method for the determination of cell number is a turbidometric measurement or light scattering. This technique depends on the fact that as the number of cells in a solution increases, the solution becomes increasingly turbid (cloudy). The solution looks turbid because light passing through it is scattered by the microorganisms present and the turbidity is proportional to the number of microorganisms in the solution. The turbidity of a culture can be measured using a photometer or a spectrophotometer. Photometers, such as the Klett-Summerson device, use a red, green or blue filter providing a broad spectrum of light.
- A **rotameter** is a device that measures the flow rate of liquid or gas in a closed tube.

19. The main roles of a baffle in a bioreactor are

- Increase aeration rate
- to prevent a vortex

20. $C_6H_{12}O_6 + \text{Zymase} \rightarrow 2\,C_2H_5OH + 2\,CO_2$ use mass balance process to calculate the ethanol production.

180g of Glucose yields = 92 g of ethanol

Therefore,

1000g of glucose yields = 92/180 * 1000 = 0.51 kg

21. The various reasons attributed to the escape of the meristems by virus invasion are:

(*a*)Viruses move readily in a plant body through the vascular system which in meristems is absent,

(*b*)A high metabolite activity in the actively dividing meristematic cells does not allow virus replication and

(c)A high endogenous auxin level in shoot apices may inhibit virus multiplication.

22. Cyanogen bromide hydrolyzes peptide bonds at the C-terminus of methionine residues. This reaction is used to reduce the size of polypeptide segments for identification and sequencing. The aspartate residue (Asp 189) located in the catalytic pocket (S1) of trypsins is responsible for attracting and stabilizing positively charged lysine and/or arginine, and is, thus, responsible for the specificity of the enzyme. This means that trypsin predominantly cleaves proteins at the carboxyl side (or "C-terminal side") of the amino acids lysine and arginine except when either is bound to a N-terminal proline. Although large-scale mass spectrometry data suggest cleavage occurs even with proline. Trypsins are considered endopeptidases, i.e., the cleavage occurs within the polypeptide chain rather than at the terminal amino acids located at the ends of polypeptides.

23. The Michaelis constant (Km) is a means of characterizing an enzyme's affinity for a substrate. The Km in an enzymatic reaction is the substrate concentration at which the reaction rate is half its maximum speed. Thus, a low Km value means that the enzyme has a high affinity for the substrate (as a "little" substrate is enough to run the reaction at half its max speed).

24. Because this enzyme has a high KM, less substrate is converted into product.

25. Final volume = initial volume + Flow rate × Number of day

$= 6m^3 + 2m^3\ day^{-1}\ (2day)$

$= 10m^3$

26. Dilution rate at end of 2nd day

$$= \frac{\text{Flow rate}}{\text{Total volume at end of } 2^{nd} \text{ day}}$$

$$= \frac{2m^3}{10m^3} = 0.2$$

27. Number of cell died

= Total cell – Number of cell survived

$= 10^9 - 10^3$

$= 9.9 \times 10^8$

$= 10^9$

28. 10^3 cpm in pellet is due to F_{ab} binding to epitope present on recombinant living cells/algae

Total F_{ab} or epitope

$$= \frac{\text{Radioactivity in pellet}}{\text{Specific activity}}$$

$$= \frac{10^3\,cpm}{10^2\,cpm(picomole)^{-1}}$$

$$= \frac{10^3 \times 10^{-12}}{10^2}$$

$= 10 \times 10^{-12}$ mole

= 10 picomole

These 10 picomole epitopes are present on 10^9 cells (assuming) 100% transfection and expression of protein).

Average number of epitopes on single recombinant alga

$$= \frac{10 \times 10^{-12}}{10^3} \text{ mole}$$

$= 10^{-14}$ mole

$= 6.023 \times 10^{23} \times 10^{-14}$

$= 6 \times 10^9$

■■

GATE-2006

BT: BIOTECHNOLOGY

Q. 1 – 6 carry one mark each

1. A thermostable DNA polymerase that can carry out both reverse transcription reaction and polymerizations has been isolated from

(*a*) Thermococcus litoralis

(*b*) Thermus aquaticus

(*c*) Thermotoga maritima

(*d*) Thermus thermophilus

2. When present in tissue culture medium, gibberellin

(*a*) helps to break dormancy of buds and bulbs

(*b*) promotes dormancy development in buds and bulbs

(*c*) is regarded as plant growth inhibitor

(*d*) prevents normal recognition of auxin molecule

3. To promote attachment and spreading of anchorage-dependent animal cells, the surface of the culture vessel needs to be coated with

(*a*) trypsin (*b*) collagen

(*c*) pronase (*d*) polyglycol

4. For amplificatio of GC rich sequences by polymerase chain reacstion, identify the reagent that binds and stabilizes AT sequences and destabilizes GC regions.

(*a*) Tetramethyl ammonium chloride

(*b*) Betaine

(*c*) 7-deaza-2′-deoxyguanosine triphosphate

(*d*) Sodium dodecyl sulphate

5. Which of the following statements is INCORRECT about immobilized plant cell cultures?

(*a*) It is possible to use high cell densities

(*b*) Cells remain active for long periods

(*c*) Cell products or inhibitors can be removed easily

(*d*) It provides low shear resistance to cells

6. All the cells that participate in immune responses originate from a population of

(*a*) neutrophils

(*b*) stem cells

(*c*) macrophages

(*d*) lymphocytes

Q. 7 – 24 carry two marks each

7. Identify the natural plant growth regulators from the following list.

(P) Zeatin

(Q) Benzylaminopurine (BAP)

(R) Indole-3-acetic acid (IAA)

(S) 2, 4-Dichlorophenoxyacetic acid

(*a*) P, Q (*b*) Q, S

(*c*) P, R (*d*) R, S

8. A hybrid derived from the fusion of a myeloma cell ($HPRT^-$) with an antibody secreting B-lymphocy te ($HPRT^+$) can be selected to produce monoclonal antibody by growing in a medium containing

(*a*) thiamine, hypoxanthine, aminopterin

(*b*) thymidine, histidine, aminopterin.

(*c*) uridine, hypoxanthine, aminopterin

(*d*) thymidine, hypoxanthine, aminopterin

9. Match items in group 1 with correct options from those given in group 2.

Group 1	**Group 2**
P. VNTR sequence	1. Gene regulation on the same chromosome
Q. Leader sequence	2. Ribosome binding site
R. SD sequence	3. DNA finger printing
S. *cis*-acting sequence	4. Functions in attenuation

(*a*) P-3, Q-1, R-4, S-2 (*b*) P-2, Q-3, R-1, S-4

(*c*) P-3, Q-4, R-2, S-1 (*d*) P-3, Q-1, R-2, S-4

10. During cultivation of microorganisms in a fermenter, various parameters are controlled by appropriate sensor (probe). Match each probe in group 1 with the appropriate response mechanism in group 2.

Group 1 (Probe)	**Group 2 (Response)**
P. Thermistor	1. Activation of acid alkali pump
Q. Oxygen electrode	2. Activation of vegetable oil pump
R. Metal rod	3. Activation of hot cold water pump
S. pH electrode	4. Increase/decrease in stirrer motor speed

(*a*) P-2, Q-3, R-1, S-4 (*b*) P-1, Q-2, R-4, S-3

(*c*) P-3, Q-2, R-4, S-1 (*d*) P-3, Q-4, R-2, S-1

11. Which of these mice fail to develop a thymus ?

(a) Motheaten mice

(b) Beige mice

(c) Knock out mice.

(d) Nude mice

12. What are the experimental steps needed for screening an expression library for a clone encoding a protein X that has been isolated and purified ?

(P) *m*-RNA isolation

(Q) Antibody preparation

(R) Cloning into an appropriate vector

(S) Western blotting on transferred plaques

(a) P, S (b) Q, S

(c) Q, R (d) R, S

13. When electroporation is used for introducing DNA into mammalian cells

(P) a carrier for DNA is not required

(Q) the lipid bilayer (membrane) interacts with an electric pulse to generate permeation sites

(R) the viability of the cells becomes approximately zero percent

(S) the first step involves absorption of DNA on the cell membrane

(a) P, Q (b) Q, R

(c) P, S (d) Q, S

14. Immobilization of enzymes using entrapment method requires

(P) photosensitive polyethylene glycol dimethacrylate

(Q) CNBr activation of sepharose

(R) polyfunctional reagent like hexamethylene diisocyanate

(S) radiation of polyvinyl alcohol

(a) P, Q (b) R, S

(c) P, S (d) Q, S

15. Which one of the following monolayer culture systems have the highest surface area: medium ratio?

(a) Roux bottle (b) Spiracell roller bottle

(c) Hollow fibres (d) Plastic bag/film

16. Majority of the cereals are highly recalcitrant to *Agrobacterium*-mediated transformation, and so direct transfor-mation methods have been developed to transform such plants. Which of the following direct transformation methods is applicable to intact plant tissues?

(a) Calcium chloride and PEG-mediated transformation

(b) Liposome-mediated transformation

(c) Electroporation

(d) Transformation using microprojectiles

17. Match items in group 1 with correct options from those given in group 2

Group 1

P. Tissue plasminogen activator

Q. Gamman interferon

R. Podophyllotoxin

S. Polyhydroxyalkanoate

Group 2

1. Immunomodulator

2. Biodegradable plastic

3. Clot dissolution

4. Anti-tumor agent

(a) P–4, Q–3, R–l, S–2

(b) P–l, Q–3, R–4, S–2

(c) P–3, Q–l, R–2, S–4

(d) P–3, Q–l, R–4, S–2

18. Match items in group 1 with correct options from those given in group 2

Group 1	Group 2
P. Amperometric biosensor	
Q. Evanescent wave biosensor	
R. Calorimetric biosensor	
S. Potentiometric biosensor	
	1. Light beam
	2. Flux of redox electrons
	3. Field effect transistors
	4. Exothermic reaction

(a) P-3, Q-4, R-2, S-1

(b) P-2, Q-l, R-4, S-3

(c) P-3, Q-2, R-4, S-1

(d) P-2, Q-4, R-3, S-1

19. For prediction of three dimensional structure of protein

(P) homology modeling tries many possible alignments

(Q) threading first identifies homologues

(R) threading evaluates many rough models

(S) homology modeling optimizes one model

(a) Q, S (b) P, Q

(c) R, S (d) Q, R

20. Immobilization of enzymes

(P) increases the specificity of the enzyme for its reactants

(Q) facilitates reuse of the enzyme in batch reactions

(R) makes it unsuitable for its use in a continuous reactor system

(S) decreases the operational cost of the industrial process

(*a*) Q, S (*b*) Q, R

(*c*) R, S (*d*) P, Q

21. Which of the following would result in somaclonal variation in micropropagated plants?

(P) Propagation by axillary branching in the absence of plant growth regulators

(Q) Cell suspension maintained for five years before induction of somatic embryogenesis

(R) Cullus induction using 20μM 2, 4-Dichlorophenoxyacetic acid followed by shoot organogenesis

(S) Shoot organogenesis from an explant in the absence of an intermediate callus phase

(*a*) P, Q (*b*) Q, R

(*c*) P, S (*d*) Q, S

22. The enzymes that can be used in 5′ end labeling of DNA are

(P) alkaline phosphatase

(Q) DNA ligase

(R) terminal transferase

(S) polynucleotide kinase

(*a*) P, S (*b*) R, Q

(*c*) P, R (*d*) R, S

Common Data Questions

Common Data for Questions 23, 24 :

Lignocellulosic biomass was subjected to microbial composting. The microbial consortium produced an extracellular enzyme xylanase, which was a glycoprotein having a molecular weight of 68 kDa and a positive charge. An aqueous extract of the enzyme could be easily prepared from the compost.

23. What techniques would you recommend for confirming the molecular weight of the purified enzyme ?

(P) Isoelectric focusing

(Q) SDS-PAGE

(R) Native PAGE

(S) Gel filtration

(*a*) P, Q (*b*) Q, S

(*c*) R, S (*d*) P, S

24. If Con A sepharose column was used for the purification of enzyme, then separation would be based on

(*a*) molecular exclusion

(*b*) affinity binding

(*c*) ion exchange

(*d*) hydrophobic interaction

Linked Answer Questions: Q. 25 to Q. 28 carry two marks each

Statement for Linked Answer Questions 25 and 26 :

DNA content of Caenorhabditis elegans was analysed and found to contain 1.0×10^8 bp.

25. How many standard λ-phage vectors carrying 20kb DNA fragments or YACs carrying 250 kb DNA fragments are theoretically required to constitute a complete C. *elegans* genomic library?

(*a*) 500 λ-phage vectors or 40 yeast clones

(*b*) 400 λ-phage vectors or 5000 yeast clones

(*c*) 5000 λ-phage vectors or 400 yeast clones

(*d*) 5×10^4 λ-phage vectors or 4000 yeast clones

26. How many λ-phage vectors/yeast clones should be prepared in order to ensure that every sequence is included in the library?

(*a*) 25×10^3 λ-phage vectors/2000 yeast clones

(*b*) 20×10^3 λ-phage vectors/1600 yeast clones

(*c*) 5×104 λ-phage vectors/4000 yeast clones

(*d*) 10×104 λ-phage/10000 yeast clones

Statement for Linked Answer Questions 27 and 28 :

A bioreactor of working volume 50 m^3 produces a metabolite (X) in batch culture under given operating conditions from a substrate (S). The final concentration of metabolite (X) at the end of each run was 1.1 kg m^{-3}. The bioreactor was operated to complete 70 runs in each year.

27. What will be the annual output of the system (production of metabolite (X)) in kg per year ?

(*a*) 55 (*b*) 3850

(*c*) 45.5 (*d*) 77

28. What will be the overall productivity of the system in kg $year^{-1}$ m^{-3} ?

(*a*) 19250 (*b*) 38.50

(*c*) 3850 (*d*) 77

ANSWERS

1. (d)	2. (a)	3. (b)	4. (b)	5. (b)	6. (d)	7. (c)	8. (d)	9. (a)	10. (d)
11. (d)	12. (b)	13. (a)	14. (c)	15. (c)	16. (b)	17. (d)	18. (b)	19. (c)	20. (a)
21. (b)	22. (a)	23. (b)	24. (b)	25. (c)	26. (c)	27. (b)	28. (d)		

EXPLANATIONS

1. TthPlus DNA polymerase is isolated from the Thermus thermophilus strain. TthPlus DNA polymerase is a single 92 kDa polypeptide showing a 52 -32 exonuclease activity but lacking 32 -52 exonuclease activity. It catalyzes the polymerization of nucleotides into double-stranded DNA in the presence of $MgCl_2$. Its efficiency has been shown more particularly on large DNA fragments up to 12 kb (using lambda phage DNA as a template). TthPlus DNA polymerase is also capable of catalyzing the polymerization of DNA using a RNA template in the presence of Mn^{2+}. The ability of TthPlus DNA polymerase to reverse transcribe at elevated temperatures (70.C) minimizes the problems encountered with strong secondary structures in RNA since they are unstable at higher reaction temperatures. Higher temperatures also result in increased specificity of primer hybridization and extension. In coupled RT/PCR assays, TthPlus is about 50-100 times more efficient than regular Taq DNA polymerase.
2. Gibberellins (GAs) are plant hormones that regulate growth and influence various developmental processes, including stem elongation, germination, dormancy, flowering, sex expression, enzyme induction, and leaf and fruit senescence.
3. Type I collagen, a fibrous protein abundant in connective tissues including tendon, ligament, dermis and blood vessel, is the major component and the primary determinant of tensile strength of the extracelluar matrix (ECM). It is widely used as a thin layer on tissue-culture surfaces to enhance the attachment and proliferation of a variety of cells including endothelial cells, fibroblasts, hepatocytes, epithelial cells and etc. In addition, collagen I can self-assemble into a 3-D superamolecular gel in vitro, making it an ideal biological scaffold to promote more in vivo-like cellular morphology and function.
4. Betaine improves the PCR amplification of GC-rich DNA sequences. Betaine reduces the formation of secondary structure caused by GC-rich regions and, therefore can be used to assist the amplification of GC rich sequences by PCR. It facilitates an isostabilizing effect on the differential thermal stability of AT and GC base pairs, which may result in a larger destabilizing effect on GC pairs and stabilzes AT rich sequence by binding with them and causing conformational change.
5. The efficiency of the production process depends on the rate of release of products rather than actual rate of biosynthesis. The immobilization process reduces biosynthetic capacity. The process does not allow easy removal of cell products or inhibitors as release of products/ by-products from cell into medium is hindered.
6. A lymphocyte is a type of white blood cell in the vertebrate immune system. Under the microscope, lymphocytes can be divided into large lymphocytes and small lymphocytes. Large granular lymphocytes include natural killer cells (NK cells). Small lymphocytes consist of T cells and B cells.
7. Zeatin is a plant hormone derived from the purine adenine. It is a member of the plant growth hormone family known as cytokinins. Zeatin was first discovered in immature corn kernels from the genus Zea. It promotes growth of lateral buds and stimulates cell division to produce bushier plants if sprayed on meristems. Again the Indole-3-acetic acid (IAA) is the most common, naturally-occurring, plant hormone of the auxin class. It is the best known of the auxins, and has been the subject of extensive studies by plant physiologists.
8. A selection medium for hybrid cell lines; contains hypoxanthine; aminopterin; thymidine. Only cell lines expressing both hypoxanthine phosphoribosyl transferase (HPRT+) and thymidine kinase (TK+) can survive in this medium. Aminopterin inhibits de novo synthesis of nucleosides, while HPRT and TK supply them from hypoxanthine and thymidine.
9. The number of repeated segments at a locus varies between individuals. Such sequence regions are called VNTR's (variable number of tandem repeats) and is representative of an individual. Thus it can be used for DNA finger printing.

   The leader sequence is the sequence at the 5' end of an mRNA molecule that is not translated into protein. It contains elements for controlling gene expression by way of regulatory elements.

   Attenuation is a proposed mechanism of control in some bacterial operons which results in premature termination of transcription and which is based on the fact that, in bacteria, transcription and translation proceed simultaneously. Translation is terminated by sequestering the Shine-Dalgarno sequence (ribosomal binding site) in a hairpin-loop structure.

Cis-acting elements are often binding sites for one or more trans-acting factors. A ribosomal binding site (RBS) is a sequence on mRNA that is bound by the ribosome when initiating protein translation and is always trans acting and hence binds with cis acting sequences.

10. Thermistors monitor temperature and as per requirements very precisely activate hot/cold water pump to heat or cool down the system. The oxygen electrode directly measures the concentration of dissolved oxygen in the liquid phase and so as to regulate the oxygen level it increases the stirrer motor speed for more aeration whereas decreases speed if oxygen concentration has to be decreased. Metal rod is used to switch on/off the vegetable oil pump as vegetable oil has defoming action but its higher concentration can as well has negative effects. Thus metal rod maintains appropriate oil concentration in the fermentor. pH electrode measures the pH of the system liquid and thus varies acid/alkali pump to increase and decrease pH respectively.

11. A nude mouse is a laboratory mouse from a strain with a genetic mutation that causes a deteriorated or absent thymus, resulting in an inhibited immune system due to a greatly reduced number of T cells. The phenotype, or main outward appearance of the mouse is a lack of body hair, which gives it the "nude" nickname. The nude mouse is valuable to research because it can receive many different types of tissue and tumor grafts, as it mounts no rejection response.

12. Required for the sandwich assay, the capture antibody binds to the antigen of interest and generates a detectable signal for quantification. This antibody is usually polyclonal in nature, and is conjugated with an enzyme (most commonly, horseradish peroxidase), so that, upon substrate addition, a colored product is formed. The density of the color can then be measured spectrophotometrically as a function of antigen concentration. The western blot (sometimes called the protein immunoblot) is a widely accepted analytical technique used to detect specific proteins in the given sample of tissue homogenate or extract. It uses gel electrophoresis to separate native proteins by 3-D structure or denatured proteins by the length of the polypeptide. The proteins are then transferred to a membrane (typically nitrocellulose or PVDF), where they are stained with antibodies specific to the target protein.

13. Electroporation, or electropermeabilization, is a significant increase in the electrical conductivity and permeability of the cell plasma membrane caused by an externally applied electrical field. It is usually used in molecular biology as a way of introducing some substance into a cell, such as loading it with a molecular probe, a drug that can change the cell's function, or a piece of coding DNA.

14. Photosensitive Polyethylene glycol dimethylacrylate is used in free radical copolymer crosslinking reactions which cross links enzyme with matrix and hence used in immobilzation of enzymes. Polyvinyl alcohol is used for formation of spherical structured beads in immobilisation process but after the process is over it needs to be degraded and for this purpose gamma radiation induced degradation of polyvinyl alcohol is carried out.

15. Hollow fibers also provide a tremendous amount of surface area in a small volume. Cells grow on and around the fibers at densities of greater than 1 X 108 per mL. Hollow fiber cell culture is the only means to culture cells at in vivo like cell densities.

16. Lipofection (or liposome transfection) is a technique used to inject genetic material into a cell by means of liposomes, which are vesicles that can easily merge with the cell membrane since they are both made of a phospholipid bilayer. Lipofection generally uses a positively charged (cationic) lipid to form an aggregate with the negatively charged (anionic) genetic material and hence is the best mechanism to transform intact plant tissues.

17. Tissue plasminogen activator is a protein involved in the breakdown of blood clots. The importance of IFNγ in the immune system stems in part from its ability to inhibit viral replication directly, and most importantly from its immunostimulatory and immunomodulatory effects. podophyllotoxin is the pharmacological precursor for the important anticancer drug etoposide.

    Polyhydroxyalkanoates or PHAs are linear polyesters produced in nature by bacterial fermentation of sugar or lipids, used in the production of biodegradable plastics.

18. Amperometric biosensors function by the production of a current generated by flux of redox electrons when a potential is applied between two electrodes.

    Evanescent wave biosensors are fiber-optic sensors which detect the refractive index of the light beams.

    Many enzyme catalysed reactions are exothermic, generating heat, this principle is used as a basis for measuring the rate of reaction and, hence, analyze concentration by calorimetric biosensors.

    Potentiometric biosensors make use of ion-selective electrodes in order to transduce the biological reaction into an electrical signal. A recent development from ion-selective electrodes is the production of ion-selective field effect transistors (ISFETs) and their biosensor use as enzyme-linked field effect transistors.

19. Protein threading, also known as fold recognition, is a method of protein modeling (i.e. computational protein structure prediction) which is used to model those proteins which have the same fold as proteins of known structures, but do not have homologous proteins with known structure. It differs from the homology modeling method of structure prediction as it (protein threading) is used for proteins which do not have their homologous protein structures deposited in the Protein Data Bank (PDB), whereas homology modeling is used for those proteins which do. Threading works by using statistical knowledge of the relationship between the structures deposited in the PDB and the sequence of the protein which one wishes to model. Homology modeling, also known as comparative modeling of protein, refers to constructing an atomic-resolution model of the "target" protein from its amino acid sequence and an experimental three-dimensional structure of a related homologous protein (the "template"). Homology modeling relies on the identification of one or more known protein structures likely to resemble the structure of the query sequence, and on the production of an alignment that maps residues in the query sequence to residues in the template sequence.

20. The immobilized enzyme is easily removed from the reaction making it easy to recycle the biocatalyst. Immobilized enzymes typically have greater thermal and operational stability than the soluble form of the enzyme. This reduces the operational cost.

21. Somaclonal variation is not restricted to, but is particularly common in, plants regenerated from callus. The variations can be genotypic or phenotypic, which in the latter case can be either genetic or epigenetic in origin. Typical genetic alterations are: changes in chromosome numbers (polyploidy and aneuploidy), chromosome structure (translocations, deletions, insertions and duplications) and DNA sequence (base mutations). Typical epigenetic related events are: gene amplification and gene methylation.

22. Alkaline phosphatase is a hydrolase enzyme responsible for removing phosphate groups from many types of molecules, including nucleotides, proteins, and alkaloids. The process of removing the phosphate group is called dephosphorylation. Polynucleotide kinase (or PNK) is a T7 bacteriophage (or T4 bacteriophage) enzyme that catalyzes the transfer of a gamma-phosphate from ATP to the free hydroxyl end of the 5' DNA or RNA.The resulting product could be used to end-label DNA or RNA, or in a ligation reaction.

23. Gel filtration is a separation based on size and same is the purpose of SDS-PAGE i.e. to separate proteins according to their size, and no other physical feature.

24. Con A has varying affinities for various polypeptides and hence affinity binding will be used.

25. Number of λ-phage vectors $= \dfrac{1.0\times10^{8}}{20\times10^{3}} = 5\times10^{3}$

Number of yeast clones $= \dfrac{10^{8}}{250\times10^{3}} = 400$

26. λ-phage vectors

$$= \frac{\text{in}(1-p)}{\text{in}(1-a/b)} = \frac{\text{in}(0.0001)}{\text{in}\left(1-\dfrac{20\times10^{3}\,bp}{10^{8}\,bp}\right)}$$

$$= \frac{\text{in}(0.0001)}{\text{in}(1-0.0002)} = \frac{\text{in}(0.0001)}{\text{in}(0.9998)}$$

$$= \frac{-9.210}{-2.0\times10^{-4}} = 5\times10^{4}$$

$$\text{Yeast clones} = \frac{\text{in}(0.0001)}{\text{in}\left(1-\dfrac{250\times10^{3}}{10^{8}}\right)}$$

$$= \frac{-9.210}{\text{in}(1-0.0025)}$$

$$= \frac{-9.210}{\text{in}(0.9975)} = \frac{-9.210}{-2.5\times10^{-3}} = 4000$$

27. In single run metabolite (X) concentration is

$=1.1\ \text{kg}\ m^{-3}$.

Since, total number of runs = 70

Total concentration of (X)

$= 70 \times 1.1\ \text{kg}\ m^{-3}\,\text{run}^{-1}$

$= 70 \times 1.1\ \text{kg}\ m^{-3}$

$= 77\ \text{kg}\ m^{-3}$

Now, total volume of batch culture = 50 m^3.

Total amount will be

$= 50\ m^3 \times 77\ \text{kg}\ m^{-3}$

= 3850 kg per year

28. Overall productivity

= Total number of runs × final concentration at each end

$= 70\ \text{runs} \times 1.1\ \text{kg}\ m^{-3}$

$= 77\ \text{kg}\ m^{-3}\ \text{year}^{-1}$

■■